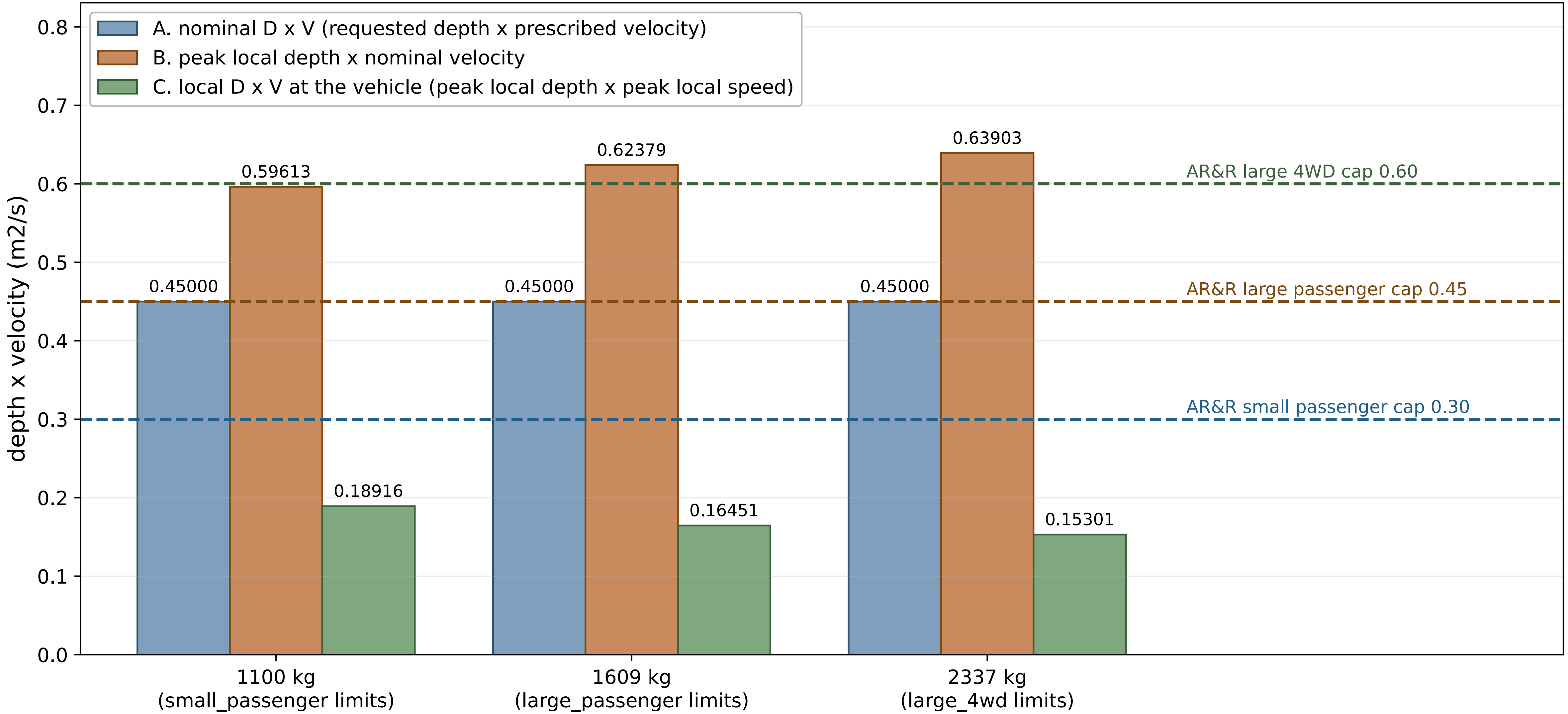
G6 Three readings of D x V from the same simulation, and they disagree



AR&R never defines D and V for a transient surge, so the same dry-start rollout supports opposite L1a verdicts depending on which reading you take. Reading A puts every run at 0.4500 m2/s. Reading B, using the bow-wave peak, pushes all three ABOVE reading A (0.5961 to 0.6390 m2/s). Reading C, measured at the vehicle where local depth rises but local speed collapses at stagnation, drops them to 0.18916, 0.16451 and 0.15301 m2/s, 58 to 66 percent BELOW nominal.

Under reading A the 1609 kg run is FORD; re-running the live L1_verdict on reading C makes it NO-FORD, and that flip is driven by the DEPTH cap, not the product cap (local peak depth 0.4159 m exceeds its 0.40 m limit while its local D x V of 0.1645 sits far under the 0.45 cap).

PRIMARY FOR THIS PROJECT: reading A, the nominal convention. That is what gates_all_runs.py feeds to vehicle_params.L1_verdict for every run in G3, and it is chosen because it is an INPUT to the simulation rather than an output of it, so a practitioner can evaluate it without running any simulation, which is the entire point of testing a cheap criterion against L2. Readings B and C depend on probe placement and so cannot be reproduced from the flood condition alone.

This is a DISCLOSURE, not a defect: the L1-versus-L2 comparison is not well posed until the criterion's own input is pinned down. Source: renders/yaris_render_s1/gates_results.json, dry-start runs m1100, m1609, m2337.